

COMMUNICATIONS TECHNOLOGY

Communications technology affects all aspects of our lives in a fundamental way. Having an understanding of communications technology is an important part of being both technologically and media literate.

Communications technology courses are project-based and will provide students with opportunities to acquire the knowledge and skills required to design, use, and manage electronic, live, recorded, and graphic communications systems, specifically in the areas of TV, video, and movie production; radio and audio production; print and graphic communications; photography; digital imaging; broadcast journalism; and interactive new media and animation. These courses will help students understand the effects of communications technology on the environment and society. Students will also examine standards and regulations governing communications technology, health and safety issues, careers in the field, and the importance of lifelong learning, and will learn about the Essential Skills and work habits that are important for success in careers in the field.

The list of approved emphasis areas for communications technology can be found at www.edu.gov.on.ca/eng/curriculum/secondary/teched.html.

- Courses in technological education are suitable for use in cooperative education programs and in connection with other forms of experiential learning as well as in programs such as the Specialist High Skills Major (SHSM). For more information, see pages 43–44 of this document.
- For policy guidelines pertaining to multiple-credit courses and emphasis courses, see pages 17–18 of this document.

Communications Technology: Digital Imagery and Web Design, Grade 12

Open

TGJ40

This course enables students to develop knowledge and skills in the areas of photography, digital imaging, animation, 3D modelling, and web design. Students will work both independently and as part of a production team to design and produce media products in a project-driven environment. Practical projects may include photo galleries, digital images, animations, 3D models, and websites. Students will also expand their awareness of environmental and societal issues related to communications technology, and will explore postsecondary education, training, and career opportunities.

Prerequisite: None

A. COMMUNICATIONS TECHNOLOGY FUNDAMENTALS

OVERALL EXPECTATIONS

By the end of this course, students will:

- A1.** demonstrate an understanding of the core concepts, techniques, and skills required to produce a range of photographic, digital imaging, animation, 3D modelling, and/or web design products or services;
- A2.** demonstrate an understanding of technical terminology, scientific concepts, and mathematical concepts used in photography, digital imaging, animation, 3D modelling, and/or web design and apply them to the creation of media products;
- A3.** demonstrate an understanding of and apply the interpersonal skills necessary to work effectively with clients and peers.

SPECIFIC EXPECTATIONS

A1. Core Concepts, Techniques, and Skills

By the end of this course, students will:

- A1.1** demonstrate an understanding of technological concepts (*e.g., aesthetics, control, environmental sustainability, ergonomics, fabrication/building/creation, function, innovation, material, mechanism, power and energy, safety, structure, systems*) and their relevance to the design and creation of media projects (see pp. 7–8);
- A1.2** use photographic, imaging, and computer equipment safely and correctly to perform basic production tasks or create simple products (*e.g., set up cameras, tripods, and lights; capture images with a digital camera or scanner; transfer images between devices*);
- A1.3** use imaging, image editing, animation, 3D modelling, and web design software correctly to perform basic production tasks or create simple products (*e.g., correcting and manipulating images, preparing images for web or print viewing, tweening, texture mapping, creating an animated GIF, designing a web page*);
- A1.4** demonstrate an understanding of the creative skills and techniques required to produce effective photographs, digital images, animations, 3D models, and/or web pages (*e.g., composition, lighting, image editing and optimization, claymation, site planning*);

A1.5 identify components of photographic and imaging systems (*e.g., cameras, camera supports and accessories, scanners, computers, monitors, printers*) and devices (*e.g., camera controls, sensor, mirror*), and describe their functions;

A1.6 use safe and appropriate methods for installing, transporting, and storing photographic and imaging equipment (*e.g., removable storage devices, cameras, cables, flashes, tripods, lights, scanners*);

A1.7 apply basic design principles (*e.g., balance, rhythm, proportion, contrast, flow*) and elements (*e.g., colour, line, space, form, texture*) to communicate an idea or concept.

A2. Technical Terminology and Scientific and Mathematical Concepts

By the end of this course, students will:

- A2.1** demonstrate an understanding of terminology related to photography, digital imaging, animation, 3D modelling, and web design, and use it correctly in oral and written communication (*e.g., resampling, rule of thirds, aperture, resolution, image map, raster and vector formats, blending, colour mode, key frames, rendering, spline, control points*);

A2.2 demonstrate an understanding of scientific concepts that relate to processes and technologies used in photography, digital imaging, animation, 3D modelling, and web design (*e.g., behaviour of light, persistence of vision, digital encoding of data, colour temperature, fibre optics, operation of image sensors*);

A2.3 use mathematical concepts and formulas as required to complete tasks in photography, digital imaging, animation, 3D modelling, and web design (*e.g., f16 rule, file size calculations, image scaling, simulation of effect of gravity on a bouncing ball in an animation, calculation of frame rates*).

A3. Teamwork

By the end of this course, students will:

A3.1 explain the value of sharing ideas, information, resources, and expertise when working in a team setting;

A3.2 describe and use techniques that encourage participation by all members of a team (*e.g., brainstorming, group discussion, celebration of others' thoughts or contributions*);

A3.3 describe and use concepts and techniques that facilitate effective collaboration in a team environment (*e.g., cooperative discussion, conflict resolution techniques, motivation techniques, respect for the ideas of others, constructive criticism*).

B. COMMUNICATIONS TECHNOLOGY SKILLS

OVERALL EXPECTATIONS

By the end of this course, students will:

- B1.** use project management techniques effectively in the development of photographic, digital imaging, animation, 3D modelling, and web design products;
- B2.** apply a design process or other problem-solving processes or strategies to meet a range of challenges in creating photographic, digital imaging, animation, 3D modelling, and web design products;
- B3.** create productions that demonstrate competence in the application of creative and technical skills.

SPECIFIC EXPECTATIONS

B1. Project Management

By the end of this course, students will:

- B1.1** use planning techniques and tools effectively (*e.g., research, storyboards, scripts, site plans, roughs, mock-ups, design briefs*) when creating plans for a photographic, digital imaging, animation, 3D modelling, and/or web design project;
- B1.2** use appropriate organizational and time-management tools (*e.g., student planners, journals, production schedules, electronic organizers, organizational software*) throughout the project to ensure that project deadlines are met.

B2. Problem Solving

By the end of this course, students will:

- B2.1** apply the steps in a design process to develop solutions to creative challenges (*e.g., define the problem or challenge, taking into account relevant contextual or background information; define project objectives and criteria and identify constraints such as cost, time, or technology limitations; gather information and generate possible solutions using techniques such as brainstorming; evaluate possible solutions and apply the one that most effectively meets the objectives and criteria within the existing constraints*) (see pp. 22–23);

- B2.2** apply appropriate problem-solving approaches and/or techniques to solve specific technical problems (*e.g., diagnostics, reverse engineering, trial and error, divide and conquer, parts substitution, extreme cases*).

B3. Process and Production Skills

By the end of this course, students will:

- B3.1** apply creative skills, equipment operating skills, and software skills to create components for a media production (*e.g., create images and animations, design page layout and navigation scheme, edit markup language for a website; shoot photographs, process images, make and mount prints for a photographic exhibition*);
- B3.2** apply editing skills to integrate the components into a unified and effective production.

C. TECHNOLOGY, THE ENVIRONMENT, AND SOCIETY

OVERALL EXPECTATIONS

By the end of this course, students will:

- C1.** describe the environmental impact of communications media technologies, and identify ways of minimizing their harmful effects;
- C2.** demonstrate an understanding of social effects and issues arising from the use of communications media technologies and the importance of respecting cultural and societal diversity in the production of media projects.

SPECIFIC EXPECTATIONS

C1. Technology and the Environment

By the end of this course, students will:

- C1.1** describe the effects of current photographic, digital imaging, animation, 3D modelling, and web design technologies on the environment (*e.g., paperless publication, increased energy consumption, battery disposal, waste and disposal problems created by rapid obsolescence of equipment*);
- C1.2** describe ways in which environmental problems are being or can be addressed by the photographic, digital imaging, animation, 3D modelling, and web design industries (*e.g., use of rechargeable batteries; reduction of packaging; recycling of paper, toner and ink cartridges; use of energy-efficient equipment; upgrading rather than replacing obsolete equipment; recycling equipment slated for disposal; environmental certification of operations [EcoLogo, ISO 14001]*).

- C2.2** identify legal and ethical issues applicable to communications media production (*e.g., copyright, respect of privacy and personal information*);

- C2.3** identify recent innovations in photography, digital imaging, animation, 3D modelling, and web design technology (*e.g., replacement of film photography by digital photography, web publication as an alternative to print publication, development of software for performing complex tasks*), and describe their social and economic effects (*e.g., decline of film manufacturing and related businesses; changes in the professional photography market; new opportunities to reach specialized or mass audiences via the Internet; increased access to information; new ways for individuals and communities to interact socially, to share information, ideas, and creative work, and to influence public opinion; rapid product obsolescence; shifting job opportunities*).

C2. Technology and Society

By the end of this course, students will:

- C2.1** demonstrate an understanding of social standards and cultural sensitivity, and use appropriate and inclusive content, images, and language in communications media productions (*e.g., by including people from different races, cultures, and backgrounds in media productions; portraying minority groups with respect and sensitivity; avoiding sexism, homophobia, and cultural or racial bias*);

D. PROFESSIONAL PRACTICE AND CAREER OPPORTUNITIES

OVERALL EXPECTATIONS

By the end of this course, students will:

- D1.** demonstrate an understanding of and apply safe work practices in the use of photographic, imaging, and computer equipment;
- D2.** identify careers in photography, digital imaging, animation, 3D modelling, and web design, and describe the skills, work habits, education, and training required for entry into employment in these fields.

SPECIFIC EXPECTATIONS

D1. Health and Safety

By the end of this course, students will:

- D1.1** describe industry hazards (*e.g., ergonomic hazards, electrical hazards, mechanical hazards*), identify sources of hazard information (*e.g., Workplace Hazardous Materials Information System [WHMIS], Passport to Safety*), and describe methods for preventing accidents;
- D1.2** demonstrate an understanding of and apply safe work practices when using equipment (*e.g., use of gloves and other protective clothing, correct seat placement, use of proper cable management techniques, use of ergonomically designed equipment, proper grounding of electrical devices, use of safety chains, correct use of ladders, proper use and storage of equipment*);
- D1.3** describe health risks (*e.g., carpal tunnel syndrome, eye strain*) associated with the use of photographic, imaging, and computer equipment, and identify ways of avoiding them.

D2. Career Opportunities

By the end of this course, students will:

- D2.1** identify career opportunities in photography, digital imaging, animation, 3D modelling, and/or web design (*e.g., photographer, animator, web designer*), and describe the qualifications and skills needed for entry into these occupations;
- D2.2** identify groups and programs that are available to support students who are interested in pursuing non-traditional career choices in the

photography, digital imaging, animation, 3D modelling, and/or web design industries (*e.g., mentoring programs, virtual networking/support groups, specialized postsecondary programs, relevant trade/industry associations*);

- D2.3** demonstrate an understanding of and apply the Essential Skills that are important for success in the photography, digital imaging, animation, 3D modelling, and/or web design industries, as identified in the Ontario Skills Passport (*e.g., computer use, oral communication, measurement and calculation, numerical estimation, job task planning and organizing, decision making*);
- D2.4** demonstrate an understanding of and apply the work habits that are important for success in the photography, digital imaging, animation, 3D modelling, and/or web design industries, as identified in the Ontario Skills Passport (*e.g., working safely, teamwork, reliability, initiative, customer service, entrepreneurship*);
- D2.5** maintain an up-to-date portfolio that includes pieces of work and other materials that provide evidence of their skills and achievements in photography, digital imaging, animation, 3D modelling, and/or web design (*e.g., work logs, skills checklist, photographs, digital media, sketches, drawings*), and explain why having a current portfolio is important for career development and advancement.